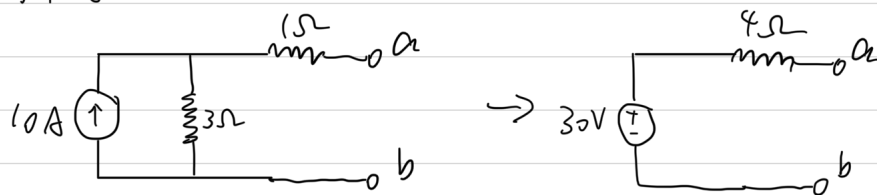


课堂作业整理 3.20

① 例 4-8

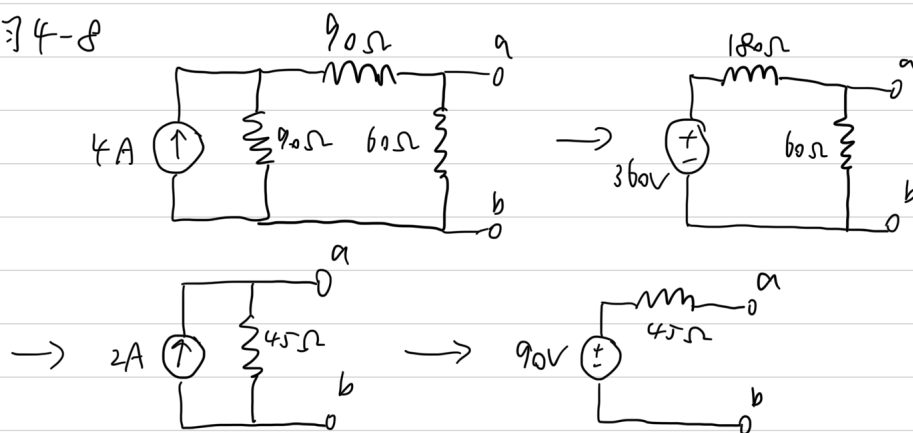


$$V_{Th} = 30V \quad R_{Th} = 4\Omega$$

$$R_L = 6\Omega \text{ 时, } I_L = \frac{30}{10} = 3A, \quad R_L = 16\Omega \text{ 时, } I_L = \frac{30}{20} = 1.5A.$$

$$R_L = 36\Omega \text{ 时, } I_L = \frac{30}{40} = 0.75A.$$

② 练习 4-8



$$\frac{180 \times 60}{180 + 60} = 45\Omega$$

$$\text{因此 } V_{Th} = 90V, \quad R_{Th} = 45\Omega \quad I = \frac{90}{15 + 45} = 1.5A$$

③ 例 4-9

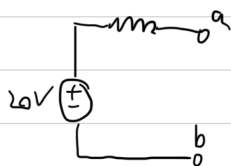
用 1V 电压源激发网络, $R_{Th} = \frac{1}{i_o}$, 对回路 1, $-2V_x + 2(i_1 - i_2) = 0$. 且 $V_x = -4i_2$

对回路 2, 3 $4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0$. $6(i_1 - i_2) + 2i_3 + 1 = 0$

得 $i_3 = -\frac{1}{6}A$, $i_o = -i_3 = \frac{1}{6}A$, 因此 $R_{Th} = \frac{1V}{i_o} = 6\Omega$

$i_1 = 1$, $4(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$, $-2V_x + 2(i_1 - i_2) = 0$, $4(i_1 - i_2) = V_x$

可得 $V_{Th} = V_{oc} = 6i_2 = 20V$



④ 练习 4-9

设在 a, b 间加 $1V$ 电压源, 电流为 I_0 由 b 流向 a

$$\text{则 } 5I_1 + 3I_x + 1 = 0$$

$$I_1 + 1.5I_x = I_x$$

$$I_x + 10 = \frac{1}{4}$$

$$\text{可得 } I_x = -2A, I_0 = 2.25A,$$

$$\text{则 } R_{Th} = 444.4 m\Omega$$

$$\text{又由 } -6 + 5 \times I_1 + 3 \times I_x + 4 \times I_x = 0$$

$$I_1 + 1.5 \times I_x = I_x$$

$$V_{Th} = I_x \times 4$$

$$4.5 I_x = 6$$

$$\text{则 } V_{Th} = 4 \times 6 / 4.5 = 5.333V$$

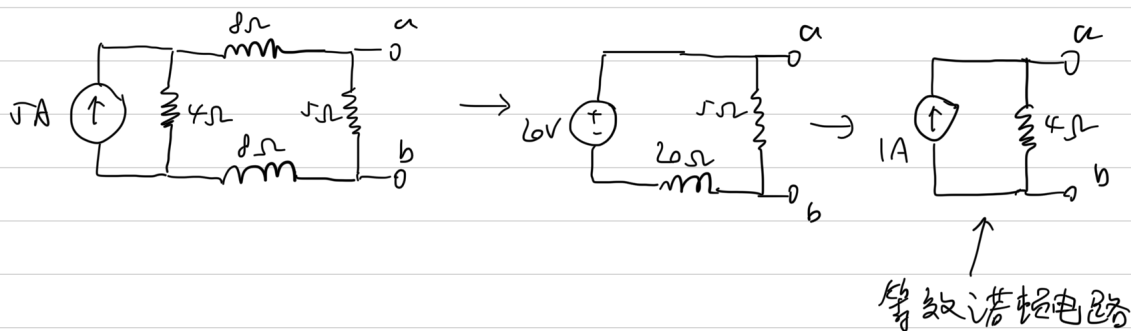
⑤ 练习 4-10

因无独立电源, 所以 $V_{Th} = 0V$, 假设 ab 间加 $1V$ 电压源, 流出的电流为

$$I_0, \text{ 方向由 } b \text{ 指向 } a, 5I_1 + 10I_1 + 4V_x + 1 = 0, V_x = -5I_1, I_1 = 0.2A,$$

$$I_1 + I_0 = \frac{1}{15}, I_0 = \frac{1}{15} - \frac{1}{5} = -\frac{2}{15}, R_{Th} = \frac{1}{I_0} = -7.5\Omega$$

⑥ 例 4-11



⑦ 练 4-11

$$R_{Th} = (90 + 90) \parallel 80 = 90\Omega$$

$$I = \frac{450}{180} + 2 = 2.5 + 2 = 4.5A$$